Project On

" KISAN SETU – *Farming Equipment Rental System.*"

"Your Farm, Our Support."

Introduction

Agriculture is the backbone of our economy ,but many small-scale farmers face challenges in affording modern and expensive farming equipment So we can created The **Kisan Setu** is a Farming Equipment Rental Management System designed to connect farmers and equipment owners through a digital platform.

The system enables farmers to rent agricultural equipment from owners at affordable rates instead of purchasing them.
It also ensures that owners can earn income by renting out idle equipment.

This system simplifies the process of renting, booking and managing farming equipment through a centralized database and digital platform.

Aim

The aim of KISAN SETU is to transform the farming equipment rental process, making it easier, faster and more affordable for farmers while ensuring fair revenue for equipment owners

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Problem Definition

Many small and medium-scale farmers cannot afford to purchase costly agricultural equipment, which leads to low productivity.
On the other hand, many equipment owners have idle machinery that goes unused for most of the year.
There is a lack of a centralized system to manage the rental process, track bookings, handle payments, and provide discounts to low-income farmers.

The problem is to create a database-driven system that organizes and automates the entire process of equipment rental and management.

Objectives

The main objectives of this project are:

- To design a **database system** for managing equipment rentals.
- To allow **farmers** to easily rent equipment online.
- To enable **owners** to add and manage their equipment.
- To provide **automated discounts** for low-income farmers.
- To maintain **transparent booking and payment records**.
- To gather **feedback** through a review system.
- To support **subscription-based plans** for regular users as well as new users.

Trigger Logic

When a new **farmer** with **annual_income < 200000 (2 lakh)** is inserted, a **10% discount** is automatically added to the discounts table for one year.

SQL QUERY

```
create trigger apply_discount_after_insert after insert on users

for each row

begin

    if new.role = 'farmer' and new.annual_income < 200000 then

    insert into discounts ( user_id , discount_percent , valid_from, valid_to)

    values (new.user_id,10.00, curdate(),date_add(curdate(),interval 1
year));

    END if;

    END $$
```

Proposed Work

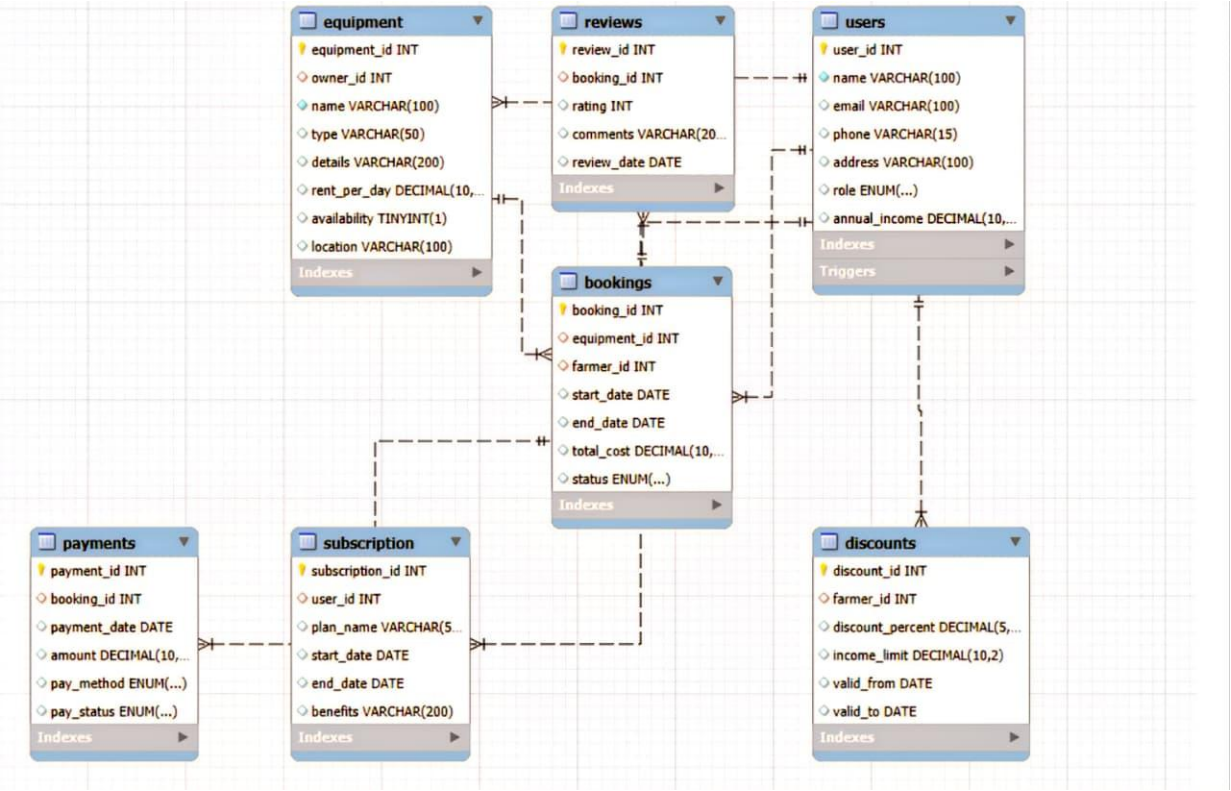
The proposed system is a **SQL-based database project** developed using **MySQL**.
It includes multiple tables and triggers to manage the workflow of equipment rental between farmers and owners.

Key Modules:

1. **Users Table** – Stores user details, roles (farmer or owner), and income information.
2. **Equipment Table** – Stores equipment details, rental prices, and availability.
3. **Bookings Table** – Manages booking details, including rental duration and total cost.
4. **Payments Table** – Tracks payment details, modes, and status.
5. **Discounts Table** – Automatically applies discounts for low-income farmers.
6. **Reviews Table** – Collects feedback and ratings for completed bookings.
7. **Subscription Table** – Manages user subscription plans and benefits.
8. **Trigger (apply_discount_after_insert)** – Automatically inserts discount details after a new low-income farmer registers.

This structure provides automation, reduces manual data entry, and ensures consistent relationships through **foreign keys**.

ER Diagram



Advantages

- Promotes **resource sharing** among farmers and owners.
- Reduces **manual data entry** and human error.
- Encourages **digital farming solutions**.
- Ensures **automatic discount management** for poor farmers.
- Provides **centralized management** for all rental activities.
- Can be **integrated with web applications** for real-time booking and payment

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Disadvantages

- **No Role based access control**
  -Even through uesrs have a role (farmer/owner),there's no backend logic restricting how can access or modify certain data.

- **Maintainance and trust issue regarding equipments**
  -There's always a risk of equipment damage or misuse. and
  I plan to include a rating/review and deposit system to reduce this risk.

- **No live gps tracking or availability updates.**
  -As of now ,owners update availability manually. future versions can use IOT or GPS.

Future Scope

- Use of **AI for predicting equipment demand**.
- **Blockchain-based transaction security**.

Conclusion

The **Kisan Setu – Farming Equipment Rental System** successfully provides a structured and efficient database design for managing equipment rentals, bookings, payments, and discounts

It bridges the gap between **farmers needing tools** and **owners willing to rent them**, promoting the concept of **"shared farming economy."**

Future enhancements could include developing a **web or mobile interface**, **AI-based rent optimization**, and **real-time analytics** for better decision making.